

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS:

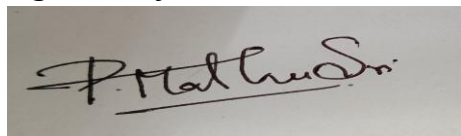
Total Marks: 20

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

Code of Conduct:

I Mathusri P-192311363 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. Define SOAP.

SOAP (Simple Object Access Protocol) is an XML-based messaging protocol used for communication between applications over a network. It provides a structured format using an **Envelope, Header, Body, and Fault**. SOAP is commonly used when secure, reliable, and standardized enterprise communication is required.

Example: A hospital system can use SOAP to exchange patient information between different departments.

2. Define WSDL.

WSDL (Web Services Description Language) is an XML-based document used to describe a web service. It specifies the **operations, input and output messages, data types, communication protocol, and service endpoint**.

In simple words, **WSDL acts as a contract between the web service and its client**.

3. What is an API?

API (Application Programming Interface) is a set of rules and interfaces that allows one software application to communicate with another application or service.

Example: A weather application can use a weather API to request current temperature data from a weather service.

4. Define XML Schema.

XML Schema (XSD) is a language used to define the **structure, elements, attributes, and data types** of an XML document.

It can specify whether a field contains a string, integer, date, or another data type. It is useful for validating XML data before processing.

Example: An XML Schema can specify that studentID must be an integer and studentName must be a string.

5. What is a Web Service?

A **web service** is a software component that allows different applications to communicate and exchange data over a network using standard web technologies.

Web services enable applications developed using different programming languages or platforms to communicate with each other.

Example: A Java application can communicate with a PHP application through a web service.

6. Define REST.

REST (Representational State Transfer) is an architectural style for designing web services using standard HTTP methods.

The common HTTP methods are:

- **GET** – retrieve data
- **POST** – create data
- **PUT** – update data
- **DELETE** – remove data

REST commonly uses **JSON** because it is lightweight and easy to process.

7. What is a Protocol?

A **protocol** is a set of rules that defines how data is transmitted and communicated between devices or applications.

Example: HTTP defines rules for communication between a web browser and a web server, while HTTPS provides secure HTTP communication.

8. Define Endpoint in Web Services.

An **endpoint** is the specific network address or URL through which a web service can be accessed.

Example:

`https://example.com/api/students`

The client sends its request to this endpoint to communicate with the service.

9. What is a Service in Web Applications?

A **service** is a software component that provides a specific functionality to other applications or clients through a defined interface.

Example: A payment service provides payment processing functionality to an e-commerce application.

10. What is the difference between SOAP and REST?

SOAP is a protocol that uses structured XML messages, while **REST** is an architectural style that commonly uses HTTP and lightweight formats such as JSON.

SOAP provides a formal messaging structure and is suitable for complex enterprise applications. REST is simpler, lightweight, and commonly preferred for web and mobile APIs.

In short:

SOAP → XML + strict structure + enterprise features

REST → HTTP + lightweight + flexible

11. What is the role of WSDL in Web Services?

WSDL describes **how a web service can be used**.

It defines:

- Available operations
- Input messages
- Output messages
- Data types
- Communication protocol
- Service endpoint

Therefore, WSDL allows a client to understand the service without knowing its internal implementation.

12. What is JSON and how is it used in APIs?

JSON (JavaScript Object Notation) is a lightweight text-based data format used to exchange structured information between applications.

Example:

```
{
  "id": 101,
  "name": "Mathu"
}
```

In APIs, JSON is commonly used as the **request and response format** because it is compact, readable, and easy for programming languages to process.

13. What is HTTP and why is it used in Web Services?

HTTP (HyperText Transfer Protocol) is a communication protocol used to transfer data between clients and servers over a network.

Web services use HTTP because it provides standard methods such as **GET, POST, PUT, and DELETE** for exchanging data.

Example:

Client → HTTP Request → Server → HTTP Response → Client

For secure communication, HTTPS is used instead of HTTP.

14. What is meant by Request and Response in Web Communication?

A **request** is a message sent by a client to a server asking for an operation or resource.

A **response** is the message returned by the server after processing the request.

For example:

Client → GET /students → Server

Server → Student Data → Client

The request may contain parameters and authentication information, while the response usually contains data and a status code.

15. What is the importance of XML in Web Services?

XML (Extensible Markup Language) is used to represent and exchange structured data.

It is important in web services because it is:

- Platform independent
- Self-descriptive
- Structured
- Human-readable
- Supported by many programming languages

SOAP uses XML as its primary message format, and XML Schema can be used to validate the data.

16. What is the purpose of an API Endpoint?

An **API endpoint** provides the specific address through which a client accesses a particular API resource or operation.

For example:

/api/students

can represent the student resource.

The endpoint helps the server identify **which service or resource the client wants to access**.

17. What is the difference between HTTP and HTTPS?

HTTP transfers data between client and server without TLS encryption.

HTTPS is HTTP secured using **TLS (Transport Layer Security)**.

HTTPS provides:

- Confidentiality
- Data integrity
- Server authentication

Therefore, HTTPS is preferred for sensitive applications such as **banking, payment systems, and login systems**.

HTTP → **Unencrypted communication**

HTTPS → **Encrypted and authenticated communication**

18. What is meant by RESTful Services?

RESTful services are web services designed according to REST principles.

They use resources and standard HTTP methods to perform operations.

For example:

GET /students → Retrieve students

POST /students → Create a student

PUT /students/101 → Update student 101

DELETE /students/101 → Delete student 101

RESTful services are generally **stateless**, meaning each request contains the information required to process it.

19. What is a Namespace in XML?

An **XML namespace** is used to uniquely identify XML elements and avoid naming conflicts between different XML vocabularies.

Example:

xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"

Here, the namespace identifies the SOAP elements and distinguishes them from application-specific XML elements.

Namespaces are especially important in **SOAP and WSDL** communication.

20. What is the role of a Client and Server in Web Services?

The **client** is the application that requests a service or data.

The **server** is the application that receives the request, processes it, and sends a response.

The basic communication is:

Client → Request → Server → Processing → Response → Client

Example: A mobile banking application is the client, while the banking web service is the server.

The client does not need to know how the server internally processes the request; it only needs to follow the service's defined interface.